

Level Views

Binary Trees, Advanced Traversals, Breadth-First Views · Mentor-Led Lab (Student Edition)

Developed by TalenciaGlobal

Difficulty ★★ Intermediate

Problem

Build a binary tree from a level-order description, then read it from the side. Produce the right-side view - the value of the rightmost node on each level, the silhouette you would see looking at the tree from the right - and the largest single-level sum.

Implement the level-order sweep and both views from scratch.

What to Compute

- Right View: the last (rightmost) node value on each level, top to bottom.
- Max Level Sum: the largest sum of values taken over any single level.

Input / Output Format

Input: One line: the tree in level order (values and N).

Output: Two lines: Right View and Max Level Sum.

Examples

Example 1

Input: tree = 1 2 3 N 5 N 4

Output:

Right View: 1 3 4

Max Level Sum: 9

Explanation: Levels are [1], [2,3], [5,4]; the rightmost of each is 1, 3, 4, and the level sums are 1, 5, 9 - the largest being 9.

Example 2

Input: tree = 10 20 30

Output:

Right View: 10 30

Max Level Sum: 50

Explanation: Two levels: [10] then [20,30]; rightmost values 10 and 30, and the largest level sum is $20 + 30 = 50$.

Constraints

- Node values are integers; N marks a missing child.
- Process the tree one full level at a time using a queue.
- The right view takes the LAST node dequeued on each level.

Sample Run

Sample Input

```
1 2 3 N 5 N 4
```

Sample Output

```
Right View: 1 3 4  
Max Level Sum: 9
```